Félix Wilhelmy

Machine Learning Engineer

Education

Master's Degree with Thesis 2024–2026
École de Technologie Supérieure
Machine Learning / Adversarial Robustness

Graduate Coursework 2025
MILA / Université de Montréal
Representation Learning (IFT6135)

Bachelor of Engineering 2018–2024
École de Technologie Supérieure
Software Engineering

DEC 2013–2016
Collège Montmorency
Computer Science

Skills

Programming Languages

Python, C++, C#, Java, SQL, TypeScript, JavaScript

Tools

PyTorch, PyTorch Lightning, NumPy, TensorBoard, Hydra, Pandas, Git, Docker, Linux, Bash, SSH, Pytest

Software Engineering

Data Structures, Algorithms, DevOps, CI/CD, Unit Testing, Software Architecture, Performance Optimization

Deep Learning

CNN, ResNet, LSTM, Transformers, LLMs, VAE, Diffusion Models

ML Development & Deployment

Data Pipelines, Model Benchmarking, Adversarial Robustness, Computer Vision, NLP, Statistical Analysis, LoRA, Self-Supervised Learning, Contrastive Learning

Languages

French (Native)

English (Professional Proficiency)

Contact

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About

I just completed my research master's in machine learning (GPA 4.06/4.30; Excellent mention at my thesis defense). I combine academic research with hands-on industrial software engineering, building complex systems from scratch and tackling open-ended technical challenges. My experience includes technical leadership and international technical competitions. A scientific paper based on my master's research is in preparation.

Experience

QB-PIER – ML Experimentation & Robustness Platform 2024–2026
Master's Research Project - École de Technologie Supérieure

- Independently built and operated an end-to-end ML experimentation platform, coordinating months of training and adversarial evaluation across 10 architectures and 50 trained model variants.
- Automated PyTorch/Lightning workflows with Hydra configurations, checkpointing, TensorBoard monitoring, and reproducible analysis of over 30 GB of experimental artifacts.

Esports Data Analyst May 2026–Sep 2026
Canadian National Team - Overwatch World Cup 2026

- Supported Team Canada's coaching staff with performance analysis, opponent scouting, and data-driven recommendations for international competition
- Built Python/SQL analytics tools and dashboards to turn game logs into player, team, and composition metrics, comparative analyses, and recurring coaching reports

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President (24–25) & VP Competitions (23–24) Sep 2023–Jan 2026
Student Club QuantumÉTS - Montréal, QC

- Led a 40-member quantum computing club, secured its first industry sponsorship, and coordinated technical workshops, competition teams, and student mentorship
- Guided teams achieving podium finishes at all six competitions attended during my presidency, including hackathons hosted by MIT, Yale, MILA, and Polytechnique Montréal

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Full Stack Developer (Internship) — React/C# Sep 2022–May 2023
Amilia - Montréal, QC

- Developed and maintained customer-facing features in a production SaaS application, including gift-card functionality across C# backend services and a React frontend
- Improved software reliability and developer workflows through unit testing, Jenkins CI/CD enhancements, debugging, and onboarding documentation

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Research Assistant (Internship) — C++/Networking Apr 2021–Aug 2021
LASSENA Research Laboratory - Montréal, QC

- Developed cross-platform C++/Qt research software for avionics analysis, including aircraft instrument interfaces and navigation-data integration
- Designed UDP-based communication for instrument data exchange and optimized background processing to maintain responsive interfaces under real-time constraints

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Full Stack Developer (Internship) — Angular/C# May 2019–Sep 2019
Air Transat - Montréal, QC

- Accelerated a critical SQL search query by approximately 6× through Entity Framework optimization, lighter query models, and asynchronous processing
- Developed internal employee flight-request and availability tools in C#/Angular; migrated Angular 7.2 to 8 and introduced lazy-loaded modules

Awards

1st Place, iQuCodeFest 2025
Polytechnique Montréal
Computing Hackathon

2nd Place, YQuantum 2025
Yale Quantum Institute
Optimization Hackathon

Best Effort Award, Mil'HaQ Fest 2025
MILA / Université de Montréal
Machine Learning Hackathon

Projects

Adaptive Graph Strategy (Hackathon) 2026
MIT iQuHack 2026 - Cambridge, MA, USA

- Co-developed an adaptive decision-making system for a stochastic graph optimization challenge, balancing expected rewards and resource constraints
- Combined Bayesian online success estimation, upper-confidence-bound exploration, and graph-scoring heuristics to refine decisions using observed outcomes

Financial Volatility Forecasting (Hackathon) 2025
Mil'HaQ Fest 2025 - Mila / Université de Montréal

- Developed a financial time-series forecasting pipeline, transforming high-dimensional swaption volatility surfaces into tractable local prediction tasks
- Implemented recurrent modeling in PyTorch with temporal feature engineering, chronological validation, and leakage-free data preprocessing

FR2SQL – French Text-to-SQL 2025
Graduate Coursework - École de technologie supérieure

- Developed a French-language Text-to-SQL prototype connecting natural-language questions to relational database schemas through semantic matching and LLM prompting
- Implemented Python/SQLite schema inspection, Sentence Transformers-based linking, Hugging Face inference components, and Spider-FR evaluation tooling

Additional Experience

Teaching Assistant Jan 2024–Aug 2025
École de Technologie Supérieure - Montréal, QC

- Supported university teaching through practical lab guidance and introductory machine-learning instruction
- Helped students implement and debug algorithms; developed technical summaries and LaTeX teaching material

Research & Development (Internship) — Machine Learning 2016
National Film Board of Canada - Montréal, QC

- Developed Python/Tesseract OCR tools supporting model training, evaluation, and research experimentation

Esports Coach 2016–2026 (Intermittent)
International Competitive Teams - Overwatch

- Coached professional and competitive esports teams internationally, leading strategic preparation, performance reviews, and player development